

Lecture 1

Finance is about decision making

Net Present value (NPV) is the difference between an investment's market value and its cost. Any positive NPV project will increase the value of the firm

$$NPV = C_0 + \frac{C_1}{(1+r)} + \frac{C_2}{(1+r)^2} + \frac{C_3}{(1+r)^3} + \dots + \frac{C_T}{(1+r)^T}$$

Future cash flows need to be discounted to find their value relative to a dollar today

Discounting accounts for the opportunity cost of the money invested: the cash flows we could have generated if we had selected the best alternative investment with identical risks.

Earnings are not cash flows:

- Accruals: revenue is recognized when sale is made not when the money is actually received
- Inventory or receivables can use or generate cash

Keep track of changes in inventory: Increases in inventory means cash outflow, decreases in inventory means selling existing product. **Corrects COGS!**

Keep track of changes in accounts receivables: Negative change in AR means cash inflow, increase in AR means not receiving cash from sales. **Corrects Sales!**

$NWC = Inventory + Receivables - Payables$

Capital expenditure (Capex) is investment in property, plants, and equipment (PP&E). Capital expenditure does not appear directly on the income statement in the year it occurs, but is a true cash outflow. Appears indirectly in subsequent years in the form of depreciation.

Depreciation is a non-cash expense! Depreciation doesn't reflect a true direct cash outflow. Impacts cash flow only indirectly through taxes

1. First, determine accounting earnings (EBIT):

$$EBIT = Revenue - COGS - SG\&A - R\&D - Depreciation$$

2. Calculates Taxes = Tax Rate x EBIT

3. Calculate Net Working Capital = Inventory + Accounts Receivable - Accounts Payable

4. Calculate annual change in NWC

5. Account for capital expenditures and add back depreciation

$$FCF = EBIT - Taxes - \Delta NWC - Capex + Depreciation$$

Investors prefer to move to a more accelerated D&A → increase NPV → promote economy

ΔNWC : correct time issues between Rev and Cos in the IS

Lecture 2

Risk and the Cost of Capital

Most firms/investments are risky. Investors are risk averse. Investors will demand a higher return to compensate them for risk

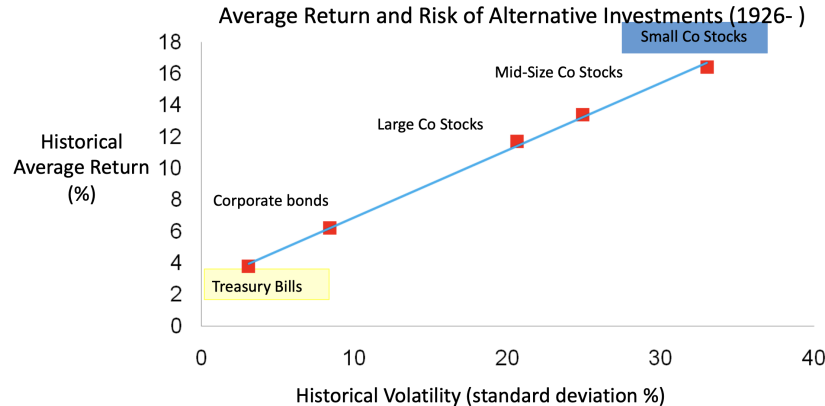
Required Return or Cost of Capital = Risk-Free + Risk Interest Rate Premium

Risk premium will depend on the amount and type of risk. Match the return of investments with "comparable" risk

Required return = minimum reward paid to investors for similar risks

Cost of capital = what the firm has to pay

Higher Risk is Associated with Higher Average Returns (Higher Cost to Firms)



When discounting the project's expected free cash flows, use discount rate that compensates investors for risk they are taking. Required return = return on other investments with similar maturity/ risk

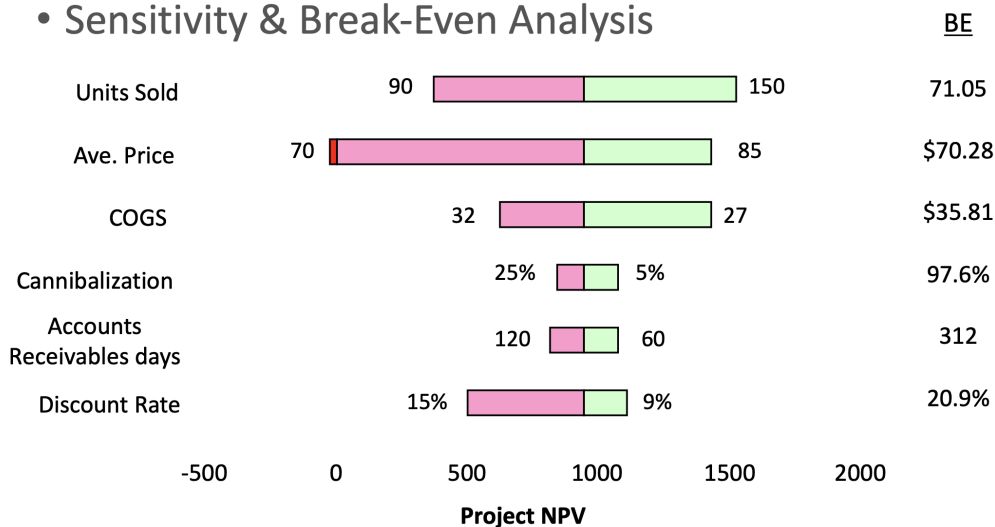
Investors demand a premium when investing in risky projects. Cash flows received sooner (rather than later) have lower discounts. Safer (vs. riskier) cash flows have lower discounts

Lecture 3

Sensitivity Analysis

We can use NPV analysis to determine the key assumptions in our forecasts

• Sensitivity & Break-Even Analysis



Internal Rate of Return (IRR)

The IRR is the discount rate at which the NPV is equal to zero. It NOT then return of the project
Calculate by using: Trial and error, Goal seek, IRR function in Excel

In the IRR excel formula include the negative first cashflow!

IRR decision rule: accept projects whose IRR exceeds a hurdle rate.

IRR > discount rate → do the project

Very common in practice: easy to communicate

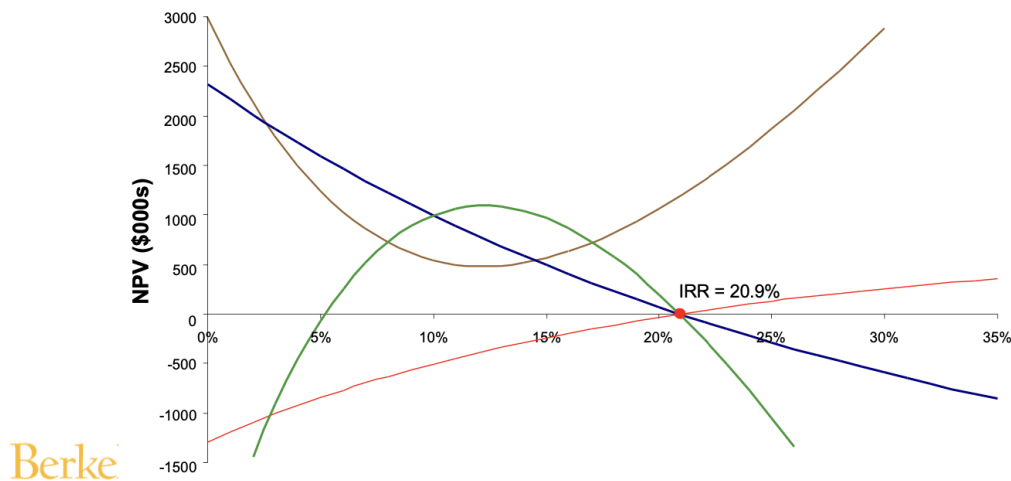
Often, hurdle rate = cost of capital

Intuition: compare benefits and costs in terms of returns

Important: Do not confuse the IRR with the opportunity cost of capital “r”. The IRR is a project characteristic while. The opportunity cost of capital is determined by the market (alternatives with similar risks)

Examples:

	Year 0	Year 1	Year 2	Year 3	Year 4	IRR()
FX-1	(3,365)	39	2,362	1,841	1,444	20.9%
Deferred Maint.	500	500	500	500	(3,290)	20.9%
Mining	(70,000)	57,250	57,250	57,250	(104,500)	5.2%
Adv. Payment	43,750	(26,750)	(50,000)	(31,500)	67,500	#NUM!



The IRR Decision Rule: Works if all negative cash flows precede positive cash flows - , - , - , + , + , + , +
In that case it coincides with the NPV decision rule when trying to accept or reject a project.

When choosing among mutually exclusive alternatives: **Choose the option with the highest NPV. Don't trust IRR!**

Lecture 4

What is inflation?

Inflation is defined as the rate of increase in prices. Inflation is typically measured by the Consumer Price Index (CPI). The CPI measures the rate of increase of the price of a bundle of goods that a typical consumer purchases.

General Principles

- The actual dollar cash flow is called the nominal cash flow.
- The actual interest rate is called the nominal interest rate.
- Relation between nominal and real cash flow:

$$C_t^{real} = \frac{C_t^{nominal}}{(1 + \pi)^t}$$

$\$C_t^{nominal}$ buys the same amount of stuff at time t as $\$C_t^{real}$ buys at time 0.

$$1 + r^{real} = \frac{1 + r^{nominal}}{1 + \pi} \quad 1 + g^{real} = \frac{1 + g^{nominal}}{1 + \pi}$$

- Fisher equation:

When faced with inflation, there are two methods for calculating PV (or FV):

- Discount (or compound) nominal cashflows with nominal interest rates.
- Discount (or compound) real cashflows with real interest rates.
- They always give the same result for the PV

Asset Sales

When a piece of capital is sold, the firm is taxed on the capital gain (or receives a deduction for the capital loss), so: **Tax from asset sale = $\tau \times (\text{Sale price} - \text{Book value})$**

Where book value is the amount of the purchase price that has not yet been depreciated: **Book value = Purchase price + Accumulated capex - Accumulated depreciation.**

Sale price versus salvage values. These are often not the same.

- Sale price: The market value you can sell something for. Changes from year to year.
- Salvage value: An accounting concept. It is the value you are depreciating the asset to. It is determined at purchase and is the expected realizable value of the asset at the end of its useful life. In US tax accounting it's generally zero.

Basic bond characteristics

Bond: a borrowing arrangement in which the borrower issues (sells) an IOU to the investor

Bond cash flows: coupons (if paid) and face value

Coupon bond: issuer makes (usually semiannual) interest payments

Face value: principal paid at maturity (and notional amount to compute interest payments)

Coupon rate: the coupon payment divided by the face value of the bond

Zero coupon bonds: no coupon payments

US Treasury bonds:

- Bills: Maturities between 1 and 12 months. No coupon payments.
- Notes: Maturities from 2 to 10 years. Coupon bonds.
- Bonds: Maturities > 10 years, up to 30. Coupon bonds.

Coupon bonds may trade at a discount, at a premium, or at par with respect to their face value:

- Price > FV: "Above Par" / "at a premium"
- Price = FV: "At Par"
- Price < FV: "Below Par" / "at a discount"

Treasury STRIPS

The Treasury does not issue long-term zero-coupon bonds directly. Since 1985, the Treasury has allowed for investors to trade individual coupons or principal payments.

STRIPS: Separate Trading of Registered Interest and Principal of Securities.

Treat each cash flow (coupon and face value) as an independent security. Treasury STRIPS are equivalent to a zero-coupon bond

It is also possible to combine STRIPS to "reconstitute" a bond. By combining STRIPS, we can replicate any other Treasury bond. This has important implications for bond prices!

Bond Yields

What is the yield on this bond? The yield is the IRR you get when holding the bond to maturity. Remember, IRR is the answer to "What rate makes the NPV of the bond = 0?"

Bond Price = PV(Future Bond Cash Flows) where the PV is calculated using the bond's yield.

Bond prices and yields move in opposite direction

Quoting Conventions

APR = Annual Percentage Rate. Interest rates are quoted on an annualized basis. A monthly interest rate (r) is quoted as $APR = r \times 12$

US bond yields are always quoted as semi-annual APRs (i.e., the IRR per 6-month period multiplied by two). Hence $YTM = 0.3450\%$ if the 6-month IRR = 0.1725%

Coupon rates are similarly quoted as semi-annual APRs.

Coupon Amount: $(\text{Coupon rate} \times \text{Face Value})/2$

Typically set so that bond will initially trade at (or close to) par

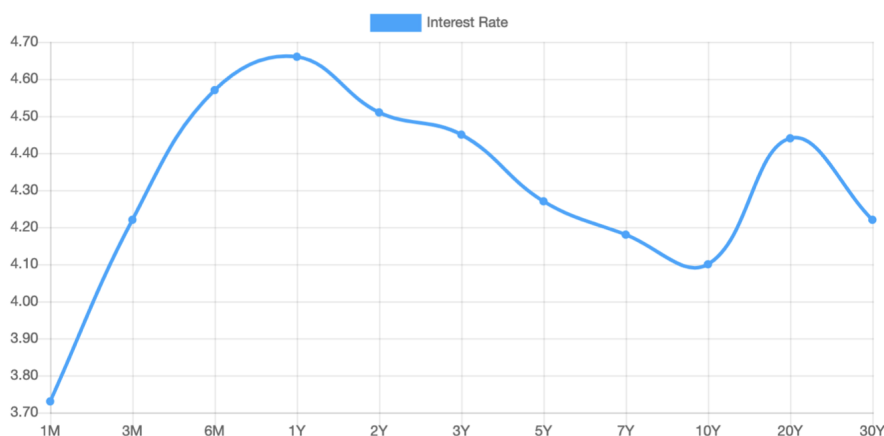
Zero Coupon Yields

Given a zero-coupon bond of price P with face value F and maturity T , the zero-coupon yield $y(T)$ is given by:

$$y(T) = \left(\frac{F}{P}\right)^{1/T} - 1$$

You can also use the IRR and RATE functions in Excel (see worksheet)

The Yield Curve



The shape of the yield curve is a very important economic indicator

- An upward sloping yield curve is an indicator that short-term interest rates will rise in the future
- A downward sloping yield curve is an indicator that short-term interest rates will fall in the future
- A downward sloping yield curve is considered a negative signal about the economy!

The shape of the yield curve changes over time

Lecture 5

Basic bond characteristics

Bond: a borrowing arrangement in which the borrower issues (sells) an IOU to the investor

Coupon bond: issuer makes (usually semiannual) interest payments

Face value: principal paid at maturity (and notional amount to compute interest payments)

Coupon rate: the coupon payment divided by the face value of the bond

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Coupon Amount: $C = \text{Coupon Rate} \times \text{Face Value} / 2$

Typically set so that bond will initially trade at (or close to) par

Bond Pricing and Arbitrage

Arbitrage: An opportunity to make money without taking on risk. It occurs whenever the same cash flows trade for different prices. Such an opportunity will be quickly exploited, and move prices so that the opportunity disappears.

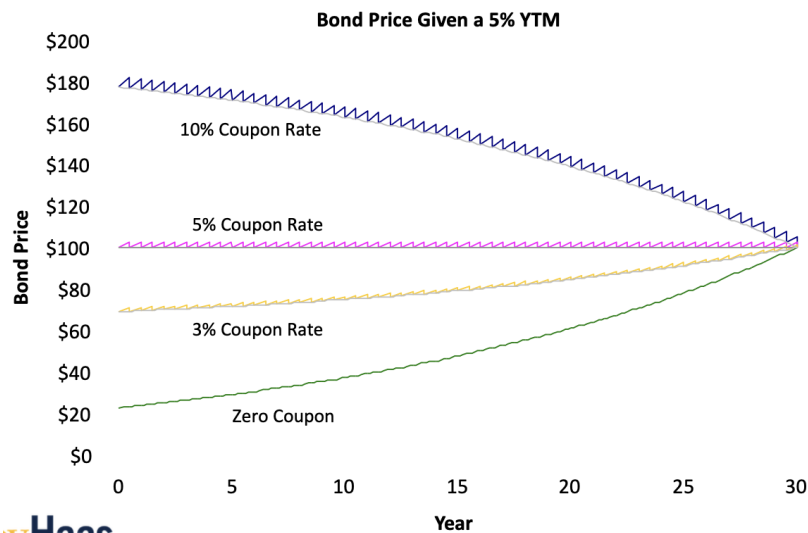
In a competitive market, arbitrage opportunities will not be present • Prices for identical cash flows are the same: “Law of One Price”

This implies that the price of a security must equal the “do it yourself” cost: $\text{Price}(\text{Security}) = \text{PV}(\text{cash flows})$ where the PV is based on competitive market prices of available securities. All other T-Bonds can be priced once the yield curve is known.

Lecture 6

Bond prices over time

As a bond matures, the price of the bond converges to its face value



Haas

Interest Rate Sensitivity of Bonds

The price of a bond is the present value of its cash flows evaluated at its YTM

As the YTM rises, the price falls. Because of discounting, this effect is most dramatic for long

As a result, even Treasury securities are not risk-free if you **do not hold** them to maturity

- If yields have fallen when you sell, your overall return will be higher
- If yields have risen when you sell, your overall return will be lower
- You will only earn the YTM if the bond's yield has not changed

The price of a t -year, zero-coupon bond will rise (fall) by about $t\%$ for each 1% drop (increase) in interest rates.

The longer the maturity of the bond, the more sensitive it is going to be to changes in the interest rate.

Example 1:

If the YTM drops to 4%:

Price change for 5yr zero = +4.9%

Price change for 10yr zero = +10.0%

Price change for 30yr zero = **+33.2%**

Example 2:

30-year zero-coupon bond

10-year 7.55%-coupon bond

100-year 7.55%-coupon bond

Which will suffer the greatest percentage loss in value if yields rise from 7.55% to 8%? **The 30-year zero-coupon bond because as it doesn't have coupons, it is more sensitive to changes in interest rate. The 10-year 7.55%-coupon bond has an effective duration of 10 years because of the coupon payments.**

Maturity Mismatch

Traditional banking follows this model: liquid short-term deposits, illiquid long-term assets (loans, mortgages). This generates maturity mismatch: Maturity mismatch occurs when assets and liabilities have different maturities.

This exposes banks to interest rate risk and the risk of self-fulfilling bank runs.

Potential Solutions

1. FDIC Insurance: Provides government backstop to bank deposits
2. Adjustable-rate mortgages (ARMs): Payment depends on the current interest rate. Also known as a floating rate security
3. Securitization (Fannie Mae, Freddie Mac): Government sponsored enterprises buy mortgages from bank balance sheets. Banks get the assets off their BS and government funds their self with these assets.

Annuities: Loans, Mortgages, etc.

An annuity provides an identical cash flow of \$C each period, starting at the end of this period and lasting for n periods.

$$PV = \frac{C}{r} \left(1 - \frac{1}{(1+r)^n} \right)$$

Suppose you take out a loan with:

- n periods
- monthly interest rate/bank discount rate r
- amount/principal P

$$C = \frac{P}{\frac{1}{r} \left(1 - \frac{1}{(1+r)^n} \right)}$$

Then loan payment C is: (annuity formula)

The annuity formula gives you the CF that makes the NPV=0 for discount rate r, or the CF that sets the PV of those CF equal to the initial loan amount.

Example "Should you pay cash?"

What is the monthly payment on the loan?

Loan rate = 14.2% APR / 12 = 1.1833% per month

Two equivalent approaches:

1. PMT Excel formula: NPER (number of periods), RATE (APR/12), PV (initial loan amount)
2. Loan payment formula

What is Mr. Sperling missing?

1. The loan balance is declining over time while the bank savings are increasing: It declines as Mr. Keppel makes his monthly payments. On average, the loan balance is half of Mr. Keppel's bank balance
2. Mr. Keppel is using his salary and not his bank savings to pay for the loan: If Keppel uses the bank savings to pay for the loan, his savings are depleted before he pays the entire loan off. The out of salary payments have an opportunity cost that Sperling is ignoring
3. He is comparing unadjusted cash-flows across periods. He assigns equal value to a dollar in interest paid after one-month and a dollar in interest received after 45 months

Lecture 7

Equity Valuation

So far, the value of an investment or bond is determined by: $\text{Price(Investment)} = \text{PV(Future Cash Flows)}$

How can we apply this to equities? Firms? What are the cash-flows? What is the right discount rate?

Alternative models:

1. Dividend discount model
2. The total payout model
3. The discounted free cash flow model
4. Multiples

Firm Valuation

Think about a firm as a collection of assets. Who owns firms' assets? Investors:

- a. Shareholders (equity): own shares (stock)
- b. Banks, bondholders (debt): own IOUs, claims on assets

What determines firm value?

1. Cash-flows: from all assets
2. Discount rates: risk-adjusted to reflect the risk of all the assets

What determines equity value?

1. Cash-flows: what is left after everyone else gets paid even debt
2. Discount rates: risk-adjusted to reflect the risk of equity

Ex: distinguishing firm from equity value

Inception:

- Firm shareholders start the firm with \$0.6m
- Firm assets \$0.6 m in cash
- **Equity value=firm value**

Firm takes a loan from the bank

- Loan \$0.4
- Firm assets \$1.0m
- **Equity value = Firm value minus debt or \$0.6m**

What happens if the firm has risky investments?

- Unpredictable cash-flows in the future.
- How do we determine equity value?

Equity: common stock

Equity (common stock) is a claim on any income remaining after all obligations including debt are paid. In other words, it is a “residual income” security. It is riskier than debt: debt is paid first!

Limited liability: equity can only lose the value of investments. Compare to unlimited liability

Common stockholder’s cash-flows:

1. Dividends: Any dividend the company chooses to pay
2. Capital gains: Price appreciation in shares

Risky Cash Flows

Cash Flows are uncertain:

- Future stock prices are obviously uncertain
- Dividends are announced shortly in advance, before that there is no commitment
- Rather than having precise estimates of these cash flows, as an investor you have “expectations” Cash Flows are expected values

Equity Cost of Capital

What is the appropriate discount rate, r_E ?

- Is buying a share of BP (BP) stock equivalent in terms of risk to investing in a one-year Treasury? No! The cash flows you receive are likely to be either higher or lower than what you expect.
- Why take this risk if you can earn the same return with no risk? Investors will require a return on the stock equal to what they expect to earn on other investments with similar risk

This return is the firm’s equity cost of capital (r_E):

r_E = risk-free rate + risk premium that depends on the stock’s risk

Each stock has a different risk profile! Stock specific risk premium

A One-Year Investor

Consider an investor who:

- buys the stock today at its current price,
- holds it for one year, receiving any dividends paid,
- sells the stock at the end of one year

Note: dividends are assumed to be paid at year-end, and P_1 is the “ex-dividend” (price w/o the dividend) stock price.

Valuation principle implies (cash-flows and discount rates):

$$\begin{array}{c}
 \begin{array}{cc}
 0 & 1 \\
 | & | \\
 -P_0 & Div_1 + P_1
 \end{array} \\
 P_0 = \frac{Div_1 + P_1}{1 + r_E}
 \end{array}
 \quad
 \begin{array}{c}
 r_E = \underbrace{\frac{Div_1}{P_0}}_{\text{Dividend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}
 \end{array}$$

Two periods: what is the stock price?

$$P_0 = \frac{D_1}{1 + r_E} + \frac{D_2}{(1 + r_E)^2} + \frac{P_2}{(1 + r_E)^2}$$

Dividend-Discount Model (DDM)

To obtain the price per share using the DDM we need: Stock price = PV(future dividends per share)

That is, the stock price should equal the present value of all future dividends

- Forecast future dividends: expected values
- Discounted using r_E , r_E compensates investors for the risk of equity in a particular firm/industry

$$P_0 = \frac{Div_1}{1+r_E} + \frac{Div_2}{(1+r_E)^2} + \dots + \frac{Div_N + P_N}{(1+r_E)^N}$$

Example: Constant Expected Growth

Special case: If dividends per share are expected to grow at a constant long run rate, g , dividends can be valued as a constant growth perpetuity:

$$P_0 = \frac{Div_1}{r_E - g} \quad \text{or equivalently,} \quad r_E = \frac{Div_1}{P_0} + g$$

Discounted Free Cash Flow (DCF)

Enterprise Value = PV (FCF) = Equity + Debt – Cash

Value of Equity = PV (FCF) + Cash - Debt

PV (FCF) = PV of cash flows from prior investments + NPV of future investments

Free Cash Flow in year t is calculated as:

$$FCF_t = EBIT_t (1 - \tau) + Dep_t - CapEx_t - \Delta NWC_t = EBIT_t (1 - \tau) - (CapEx_t - Dep_t) - \Delta NWC$$

$$EV = PV(FCF) = \sum_{t=1}^{\infty} \frac{FCF_t}{(1+r)^t}$$

Cashflows: explicit forecasting:

- Takes advantage of specific knowledge about future FCF (value drivers)
- Allows incorporation of changing investment opportunities, etc
- No need to directly consider dividend policy or financing

Discount rate: Important:

- We are discounting the FCFs that will be paid both to debt and equity holders. Thus, we discount using the firms' weighted average cost of capital (WACC) or r_{WACC} . If the firm has no debt, $r_{WACC} = r_E$.

DCF in practice:

$$V_0 = \frac{FCF_1}{1+r} + \frac{FCF_2}{(1+r)^2} + \dots + \frac{FCF_{10}}{(1+r)^{10}} + \frac{V_{10}}{(1+r)^{10}}$$

Split valuation. Two steps: (a) forecasting period, (b) terminal value

Continuation or Terminal Value:

- Beyond a 5 to 10-yr horizon, our forecasts are unlikely to be precise
- Strong tendency of firms to mean-revert to industry norms
- Use a "multiples" approach to estimate value beyond the forecast horizon, based on expected long-run growth rate. Terminal multiple is a statement of the future profitability of the firm and its industry

$$V_{10} = \frac{FCF_{11}}{r - g_{FCF}} = FCF_{10} \times \frac{1 + g_{FCF}}{r - g_{FCF}} \quad \leftarrow \text{"terminal multiple"}$$

Multiples Valuation Procedure

- Step1: Identify firms that are considered to be the same as the firm you wish to value, the comparable firms.

- Step2: Determine relevant financial ratios for the comparable firms.
- Step3: Calculate average of the financial ratios for the set of comparable firms.
- Step4: Review and modify the set of comparable companies as necessary. Analyze the subject and comparable companies to compare and contrast them in terms of both financial and economic characteristics.
- Step5: Multiply multiple by value of the characteristic for the firm you wish to value to get the value of the firm of interest.

Which step is the most difficult? Step 1! Determining the universe of comparable firms is very difficult.

Are you looking for comparable firms in terms of operations? Or financial aspects? Your business and industry knowledge will help you choose.

- Highly subjective
- How do you assess multiples when companies have multiple divisions?
- Difficult to assess brand equity and management expertise.
- Companies can be of substantially different sizes.
- The number of comparable companies to include may not be clear.
- It is very difficult to find “pure” comparable companies.

Structure of Multiples

Framework for valuation with multiples:

- Initially, you calculate a particular financial ratio “j” for some publicly traded company “i” (or private transaction), e.g. P/EPS ratio for IBM:

$$\text{Financial Ratio } j \text{ for Comparable Firm } i = \left(\frac{\text{Numerator}_i}{\text{Denominator}_i} \right)$$

- This financial ratio is calculated for a set of firms that are assumed to be comparable to the firm being valued. The “multiple” is then the average of this financial ratio “j” across the set of comparable companies.

$$\text{Multiple}_j = \text{Average} \left(\frac{\text{Numerator}_i}{\text{Denominator}_i} \right)$$

Common Multiples

a) Cash-flow based total market value (V or A) or enterprise multiples (EV): Total firm performance multiples (for all capital holders):

- V/Sales or EV/sales
- V/EBIT(DA) or EV/EBIT(DA)
- V/FCF or EV/FCF

b) Cash-flow based stock market value (P or S) multiples: Equity performance multiples (only for equity holders)

- P/Earnings per Share
- S/Earnings
- S/EBITDA
- S/FCF

c) Book value (BV) (shareholders’ equity) based multiples: Enterprise and equity performance multiples

- V/BV
- EV/BV
- S/BV

d) Industry-specific value multiples:

- EV/HospitalBeds
- EV/NumberOfCustomers

- EV/NumberofMBAs

Logic behind Multiples (comp valuation)

If Firm A and Firm B are truly comparable in all other respects, then Firm B should be priced proportionately ☹

Law of one price

But are firms really comparable in every way?

Value drivers: sales growth rate, operating profit margin, tax rates, CAPEX requirements, working capital requirements, price relative to cost, etc? Of course not!

Multiples: Common Steps

1. Pick close comparables (industry, etc) What are the fundamental value drivers in the relevant industry? Beware of differences across firms
2. Be careful about “abnormal” values / firms. Find mean, median values. Are your numbers explained by “extreme” comps?
3. Consider several multiples. Remember that multiples are often self-serving
4. Obtain an estimate of equity value. Adjust values for differences

Valuation by Comparables

Advantages

- Don’t need to estimate cost of capital, future growth rates, profit margins, etc.
- Reflects current market conditions: opportunity costs of assets
- Provides a “reality check”

Disadvantages

- No way to incorporate unique aspects of a given firm, special circumstances, potential operational improvements, etc.
- Lack of precision
- Relies on other firms being similar (twins) and “correctly” valued
- Provides no information regarding target growth rates or other aspects of future performance

Lecture 8

Interco Case

If there is no CAPEX neither D&A it is because they are investing the same that they are depreciating. Bad assumption if we are going to increase sales in 8% growth each year, they would need to increase Fixed Assets more than the increase in D&A.

NWK 0 the first 5 years: Growth Sales without increasing inventories ☹ not realistic because this is a very inventory intensive company

☹ Very aggressive the estimated DCF

Discount rate = risk-free + risk premium = 9% (long term treasury yield)+ 6% (investing in Interco is very similar to invest in the S&P 500 and the S&P 500 return is 6%)= 15%

• Terminal Multiple

$$V_{10} = \frac{FCF_{11}}{r-g} = \left[\frac{1+g}{r-g} \right] \times FCF_{10}$$

		Long Run Growth Rate					
		3%	4%	5%	6%	7%	8%
Cost of Capital	9%	17.2	20.8	26.3	35.3	53.5	108.0
	10%	14.7	17.3	21.0	26.5	35.7	54.0
	11%	12.9	14.9	17.5	21.2	26.8	36.0
	12%	11.4	13.0	15.0	17.7	21.4	27.0
	13%	10.3	11.6	13.1	15.1	17.8	21.6
	14%	9.4	10.4	11.7	13.3	15.3	18.0
	15%	8.6	9.5	10.5	11.8	13.4	15.4
	16%	7.9	8.7	9.5	10.6	11.9	13.5
	17%	7.4	8.0	8.8	9.6	10.7	12.0
	18%	6.9	7.4	8.1	8.8	9.7	10.8

$g=4\%$ and cost of capital= 15% → terminal multiple: 9.5x, and they are using 16x

Considering cost of capital of 13% they would need a g between 6% and 7% to reach a terminal multiple of 16x → pretty high

Board didn't want to lose their position and paid Wasserstein Perella to have the valuation that they wanted. Nothing in the DCF made sense.

This is an example where management chooses their friends at the BOD to comply with their wishes, which is why they were later sued by the shareholders.

Portfolio Choice: Risk and Return

Probability distribution

When projects are risky, there are different returns that a given project may earn:

- Each possible return has some likelihood of occurring (ex. beliefs)
- This information is summarized with a probability distribution, that assigns a probability, P_R , that each possible return, R , will occur.

Expected Return

Given the probability distribution of returns, we can calculate the expected return of an investment. The expected or mean return is calculated as a weighted average of the possible returns, where the weights correspond to the probabilities that each return is obtained.

Example: What is GE's expected return next year?

$$E[R] = 0.2 \times 30\% + 0.6 \times 10\% + 0.2 \times (-10\%) = 10\%$$

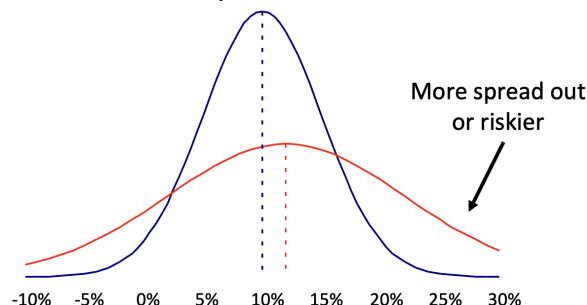
$E[R]$ refers to the expected return, often shown as R

Measures of risk

Another important characteristic of an investment is how "spread out" or risky the probability distribution of returns is.

The range of possible outcomes around an investment's expected value is related to its risk:

- Low risk: considerable "bunching" of returns around the expected value
- High risk: less clustering of returns around the expected return



The expected returns of these distributions are 10% and 12%, respectively

Variance and standard deviation

The variance and the standard deviation are two measures of how “spread out” or risky a distribution of returns is.

The variance is defined as the expected squared deviation from the mean

$$\text{Var}(R) = \sigma_R^2 = E[(R - \bar{R})^2]$$

The standard deviation is the square root of the variance.

$$\text{Std}(R) = \sigma_R = \sqrt{\text{Var}(R)}$$

The standard deviation is also referred to as volatility

Both are measures of risk!

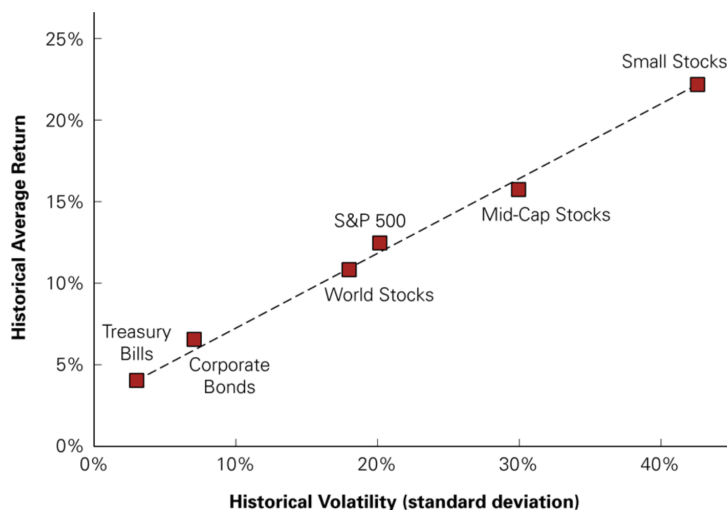
Example: What is the standard deviation (volatility) of GE?

$$\begin{aligned}\text{VAR}(R) &= [(0.2)(0.3 - 0.1)^2 + (.6)(0.1 - 0.1)^2 + (0.2)(-0.1 - 0.1)^2] = [(0.2)(0.2)^2 + (.6)(0)^2 + (0.2)(-0.2)^2] \\ &= [(0.2)(0.2)^2 + (.6)(0)^2 + (0.2)(-0.2)^2] = 0.016\end{aligned}$$

$$\text{SD}(R) = \text{SQRT}(\text{VAR}) = 0.126$$

What happens to GE’s volatility if the range of outcomes becomes more spread out? Volatility increases!

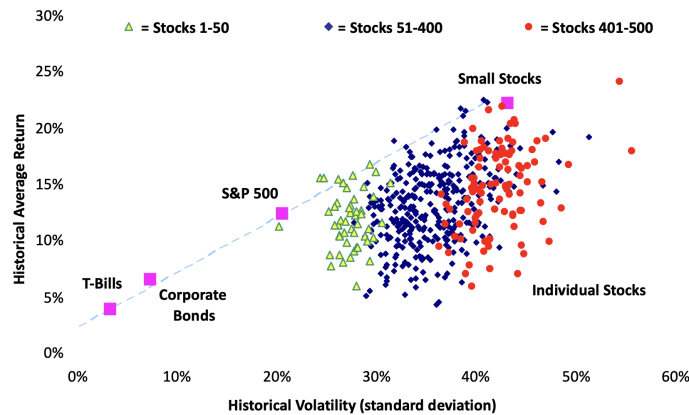
Risk vs. Return



World stocks are less risky because of the diversification

Lessons from history

1. Risky investments, on average, earn a risk premium: there is a reward for bearing risk
 2. The greater the potential reward from a risky investment, the greater the risk
- Implication: stocks with higher volatilities should have higher average returns!!! NO!!!



Volatility does not explain the returns of individual stocks

S&P 500 is not in the middle, because of the diversification, the risk is lower. When I invest in individual stocks the risk is not the same and it is higher.

Example:

	Weather	Probability	Return	Expected Risky? Return
Beach Resort	Sunny	0.50	40.0%	10%
	Rainy	0.50	-20.0%	
Umbrella Factory	Sunny	0.50	-20.0%	10%
	Rainy	0.50	40.0%	
Portfolio	Sunny	0.50	10.0%	10%
1/2 Beach Resort	Rainy	0.50	10.0%	
1/2 Umbrella Factory				

1. As long as firms' outcomes do not move in lockstep, diversification reduces risk exposure

- In the example, outcomes are perfectly negatively related: risk is eliminated!
- Diversifiable risks do not need compensation: they do not affect the cost of capital

2. Yet, some risks are not diversifiable: ex. world recessions. Those risks cannot be eliminated. Un-diversifiable risk exposure needs to be compensated!!

Portfolio Risk: The Intuition

The amount of diversification that occurs in a portfolio will depend upon the amount of risk that is common, versus the amount that is independent

Independent risks will be diversified in a large portfolio.

Common, or correlated, risks will remain in a large portfolio

How does this apply to stocks? Generally, stock prices fluctuate for two reasons:

- Good or bad news specific to the company (ex. a fire in a plant, etc)
- Good or bad news about the economy (ex. recession, etc)

By holding a large portfolio, investors can diversify the first type of risk, often called **firm-specific** or **diversifiable risk**.

The second type of risk remains even in a large portfolio: it is often called **systematic** or **market risk**.

Investors care about the correlated, market risk in their portfolios. Competition among investors implies that diversifiable risks do not affect the returns they demand for investing in a project

Idiosyncratic risk examples: company scandals, CEO death, inventories issues

Systematic risk examples: recession, nuclear war, inflation, climate changes, tax policy, oil prices

Groceries ☐ low beta because they don't suffer too much from recessions.

Diversification: summary

1. For determining expected returns, the volatility of the return of a stock is NOT the right way to measure risk. Historical evidence shows that returns and standard deviations have no clear relationship
2. Investors require compensation for risks that cannot be diversified away by forming portfolios. Investors hold diversified portfolios. Investors care about the correlated, market risk in their portfolios. To find the risk of a portfolio, we need to know the degree to which stocks returns move together

Portfolio Choice: Efficient Frontier

Covariance and Correlation

Covariance between returns of stocks A and B: $Cov(R_A, R_B) = E[(R_A - \bar{R}_A)(R_B - \bar{R}_B)]$

Captures whether stocks move together or opposite

Correlation rescales covariance to be between -1 and +1:

$$Corr(R_A, R_B) = \frac{Cov(R_A, R_B)}{Std(R_A) Std(R_B)}$$

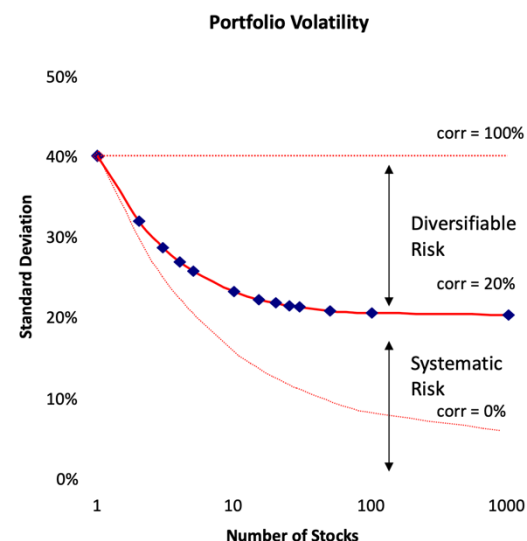
Correlation between stocks and diversification

Correlation Coefficient	Effect of Diversification (Portfolios) on Risk Reduction
+1.0	No risk reduction
0.5	"Moderate" risk reduction
0.0	"Significant" risk reduction
-0.5	"Most risk" can be eliminated
-1.0	All risk can be eliminated

Insight: any investment less than perfectly correlated with others can reduce risk if held in a portfolio.

Correlation is crucial to understand whether an investment can reduce investors' risk exposure

Diversification in practice



Consider a portfolio of stocks chosen at random

- Typical volatility = 40%
- Invest an equal amount in each stock

What happens to the volatility of the portfolio as the number of stocks in the portfolio grows?

- Depends on the correlation between stocks
- Typical correlation ~20%

When you have positive correlation, you eliminate the idiosyncratic risk with diversification, but you have still exposure to the systematic risk.

Historical Return in Excel

Average monthly return: In Excel =AVERAGE(B2:B45)=1.75%

Return APR: Annual Percentage Rate (APR) =Average monthly return *12=21.02%

Standard Deviation in Excel

Standard deviation of monthly returns

- =STDEV(B2:B45)=4.7%
- =SQRT(VAR((B2:B45))=4.7%

Japan Standard deviation APR

- =SQRT(VAR(B2:B45)*12)=16.3%
- =STDEV(B2:B45)*SQRT(12)=16.3%

Correlation US and Japan

Correlation: =CORREL(B2:B45,C2:C45) =0.349

Describing portfolios

Given a portfolio of stocks we need to be able to determine its: (a) expected return and (b) volatility

1. To determine the expected return we need:

- The fraction that is invested in each stock or what is referred as the portfolio weights of each stock. The sum of portfolio weights is equal to one (100% of your investments)
- The expected return of each stock

2. To determine its volatility we need:

- The variance of each stock, the portfolio weights, and
- The covariance between stocks or equivalently, the correlation between stocks, and the volatility of each stock

$$\text{Corr}(R_A, R_B) = \frac{\text{Cov}(R_A, R_B)}{\text{Std}(R_A) \text{Std}(R_B)} \quad \text{Cov}(R_A, R_B) = \text{Corr}(R_A, R_B) * \text{Std}(R_A) * \text{Std}(R_B)$$

Portfolios: expected return and volatility

Expected Return of a Portfolio: $R_p = x_1 R_1 + x_2 R_2 + \dots + x_n R_n = \sum_i x_i R_i$

The expected return of a portfolio is the weighted-average of the expected returns of the stocks in the portfolio, where the weights (x's) correspond to portfolio weights

Standard Deviation or volatility of a Portfolio

For a two-stock portfolio:

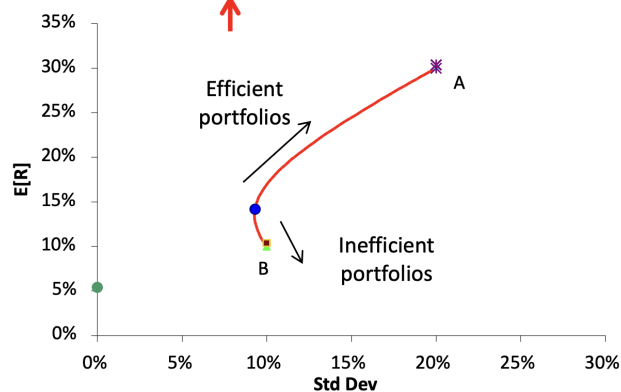
$$\begin{aligned} \text{Var}(R_p) &= x_A^2 \text{Var}(R_A) + x_B^2 \text{Var}(R_B) + 2x_A x_B \text{Cov}(R_A, R_B) \\ &= x_A^2 \text{Var}(R_A) + x_B^2 \text{Var}(R_B) + 2x_A x_B \text{Corr}(R_A, R_B) \text{Std}(R_A) \text{Std}(R_B) \end{aligned}$$

$$\text{Std}(R_p) = \sqrt{\text{Var}(R_p)}$$

2-Stock Portfolios with diversification gains

	Exp. Return	Std Dev	Corr	TBill Rate
Stock A	30%	20%		
Stock B	10%	10%	10%	5.00%

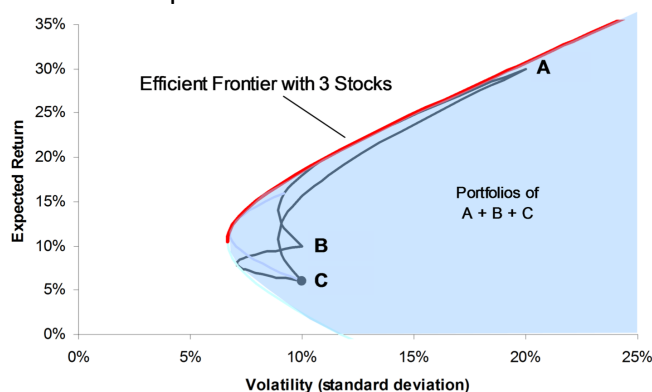
X_A	X_B	Std(R_P)	$E[R_P]$
100%	0%	20.0%	30.0%
90%	10%	18.1%	28.0%
80%	20%	16.3%	26.0%
70%	30%	14.6%	24.0%
60%	40%	13.0%	22.0%
50%	50%	11.6%	20.0%
40%	60%	10.5%	18.0%
30%	70%	9.7%	16.0%
20%	80%	9.3%	14.0%
10%	90%	9.4%	12.0%
0%	100%	10.0%	10.0%



Some portfolios are inefficient: possible to increase $E[R]$ without increasing the volatility

Many Stocks

By adding more stocks, we increase diversification opportunities, and improve the efficient frontier:
3-stock example. Note: No individual stock is on the efficient frontier!



Short Sales

Short Selling a Stock: A “short sale” is when you sell a stock that you do not own. A negative investment in a security: you sell a stock that you do not own and then buy that stock back in the future.

It is a way to bet on the stock dropping, rather than rising

How it works:

- Your broker borrows the stock from someone who owns it. The stock is then sold and you receive the proceeds
- During the time your short position is “open” you must pay the owner any dividends he would have received
- At some time in the future you will “close” your short position by buying back the stock and returning it to the person who lent it to you
- Cash flows are the reverse of buying the stock:

	0	1
Buy the stock	Pay P_0	Receive $Div_1 + P_1$
Short the stock	Receive P_0	Pay $Div_1 + P_1$

Portfolios

How do we include short sales in a portfolio?

For a short sale, we use a negative portfolio weight to represent the amount short sold (as a percentage of our total investment). Again, portfolio weights must sum to 1, all formulas work as before

Example: Suppose you have \$200 to invest, $P_A=50$, $P_B=100$, you want to short one share of B and invest all your money in A. What are the portfolio weights and the expected return of your portfolio?

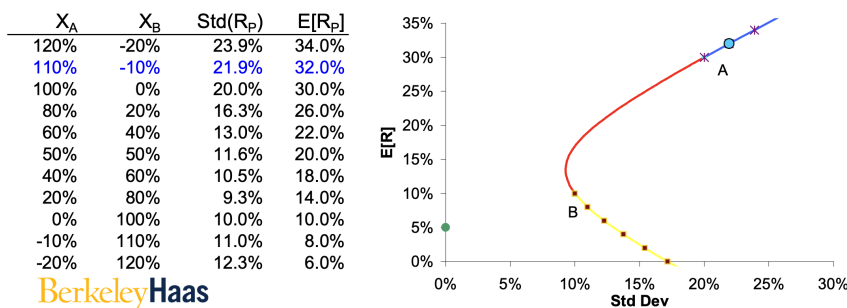
$$x_A = 300/200 = 150\%, x_B = -100/200 = -50\%$$

$$E(R_p) = x_A E(R_A) + x_B E(R_B) = 1.50 \cdot (30\%) - 0.50 \cdot (10\%) = 40\%$$

Two-stock Portfolio with Short Sales

Allowing short sales increases the set of feasible portfolios: We can increase our expected return by shorting the stock that is expected to under-perform

	Exp. Return	Std Dev	Corr	TBill Rate
Stock A	30%	20%		
Stock B	10%	10%	10%	5.00%



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Risk-free investments and leverage

Risk-free investments: Risk can also be reduced by investing a portion of a portfolio in a risk-free investment

- Treasury bills
- T-Bills pay a known return: zero volatility
- Note that a risk-free investment is also uncorrelated with any other investment

However, investing in T-bills will also reduce the expected return

Leverage: Aggressive investors seeking high expected returns may decide to borrow money to invest more than 100% of their wealth in the stock market. Higher risk, higher expected return

Invest x in a portfolio of stocks and $1-x$ in T-Bills

$$\text{Var}(xR_p + (1-x)R_f) = x^2 \text{Var}(R_p) + (1-x)^2 \text{Var}(R_f) + 2x(1-x) \text{Cov}(R_p, R_f) = x^2 \text{Var}(R_p)$$

$$\text{Std}(xR_p + (1-x)R_f) = x \text{Std}(R_p)$$

Risk is proportional to the fraction invested in stocks. Note that $x > 1$ in this case corresponds to borrowing: buying stocks on margin or borrowing money!

Risk-free investments and leverage

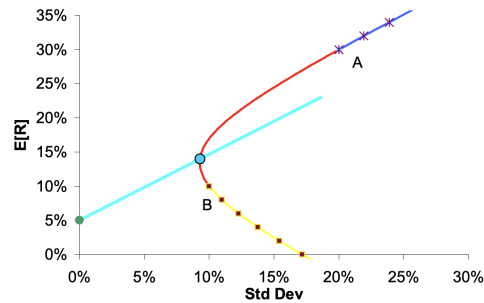
Combining with a risk-free investment (T-Bills)

Positive investment in T-bills reduces risk but also the expected return. Shorting T-bills (leverage) increase risk (but increases expected return)

	Exp. Return	Std Dev	Corr	TBill Rate
Stock A	30%	20%		
Stock B	10%	10%	10%	5.00%

X_A	X_B	Std(R_P)	$E[R_P]$
120%	-20%	23.9%	34.0%
110%	-10%	21.9%	32.0%
100%	0%	20.0%	30.0%
80%	20%	16.3%	26.0%
60%	40%	13.0%	22.0%
50%	50%	11.6%	20.0%
40%	60%	10.5%	18.0%
20%	80%	9.3%	14.0%
0%	100%	10.0%	10.0%
-10%	110%	11.0%	8.0%
-20%	120%	12.3%	6.0%

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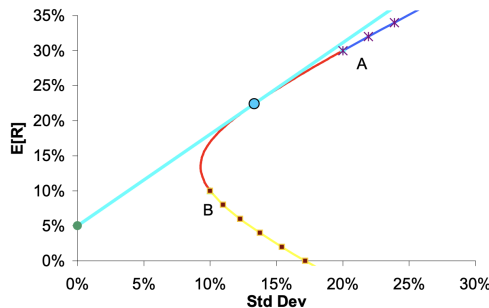


Optimal Portfolio

There is a unique **optimal** portfolio: On the line tangent to the efficient frontier of the two stocks This portfolio is optimal, independent of your risk preference

X_A	X_B	Std(R_P)	$E[R_P]$
120%	-20%	23.9%	34.0%
110%	-10%	21.9%	32.0%
100%	0%	20.0%	30.0%
80%	20%	16.3%	26.0%
62%	38%	13.3%	22.4%
50%	50%	11.6%	20.0%
40%	60%	10.5%	18.0%
20%	80%	9.3%	14.0%
0%	100%	10.0%	10.0%
-10%	110%	11.0%	8.0%
-20%	120%	12.3%	6.0%

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Sharpe Ratio

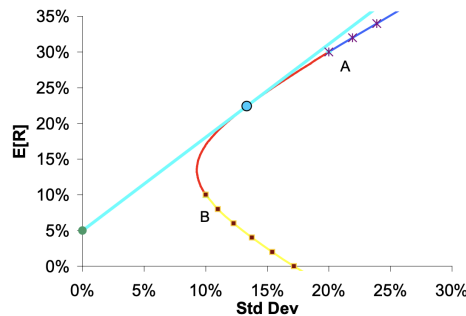
The optimal portfolio is on the line with highest slope, or Sharpe Ratio

$$\frac{E[R_P] - r_F}{\text{Std}(R_P)}$$

Sharpe Ratio = Expected Excess Return / Volatility =

X_A	X_B	Std(R_P)	$E[R_P]$	Sharpe Ratio
120%	-20%	23.9%	34.0%	1.214
110%	-10%	21.9%	32.0%	1.232
100%	0%	20.0%	30.0%	1.250
80%	20%	16.3%	26.0%	1.287
62%	38%	13.3%	22.4%	1.306
50%	50%	11.6%	20.0%	1.291
40%	60%	10.5%	18.0%	1.242
20%	80%	9.3%	14.0%	0.968
0%	100%	10.0%	10.0%	0.500
-10%	110%	11.0%	8.0%	0.273
-20%	120%	12.3%	6.0%	0.082

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In practice, how can we identify the tangent portfolio? Who Ya Gonna Call? **CAPM!**

Summary

Key Assumptions:

- Investors are risk-averse
- They evaluate portfolios based on their expected return and volatility
- Investors can lend or borrow at the risk-free interest rate

Then:

- The optimal portfolio for all investors is the tangent portfolio plus lending or borrowing
- Amount of lending/borrowing determined by risk preferences

Capital Asset Pricing Model (CAPM)

Suppose:

- All investors have similar information and expectations:
 - Same expectations about returns
 - Same estimates of covariances/correlations between returns
- All investors choose mean-variance efficient portfolios

Then: Each investor will compute the same tangent portfolio. Aggregate Demand for Stocks = Tangent Portfolio

In equilibrium, Supply = Demand:

Tangent Portfolio = Aggregate Supply of Stocks = Market Portfolio: value weighted

Examples: = S&P 500 (500 large-cap US stocks)

= Wilshire 5000 (Total US stock market)

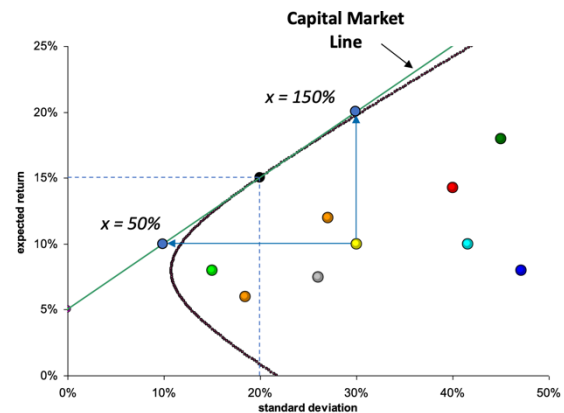
Lecture 10

The Capital Market Line (CML)

- Under the CAPM, the best possible risk-return possibilities are combinations of risk-free savings and the Market Portfolio
- Portfolios on the CML are determined by the **fraction x** invested in the market

$$\text{Exp. Ret} = r_f + x(\bar{R}_{Mkt} - r_f)$$

$$\text{Volatility} = x \text{Std}(R_{Mkt})$$



On average, active money managers do not beat the market!

Beta and Expected Return

The market portfolio is the “benchmark” portfolio:

- The “best” deal: highest Sharpe ratio
- Fully diversified market portfolio

If the CAPM holds, we can use the market portfolio to compute betas for specific projects/firms: Betas capture how costly it is to provide financing to a project, given the best alternatives in the market. Before:

$$E[R_i] = r_f + \beta_{i,market} * (E[R_{market}] - r_f)$$

$$\beta_{i,market} = \text{Corr}(R_i, R_{market}) * \frac{\text{Std}(R_i)}{\text{Std}(R_{market})} = \frac{\text{Cov}(R_i, R_{market})}{\text{Var}(R_{market})}$$

CAPM:

Idea: sensitivity to market risk

Expected percent change in the excess return of a security for a 1% change in the excess return of the market portfolio

Beta differs from volatility: volatility captures **total** risk (diversifiable and common risk), while beta is a measure of **only common** risk

Intuition: Captures the sensitivity of the underlying revenues and cash flows to general economic conditions.

Stocks in cyclical industries are likely to be more sensitive to common risk and have higher betas than stocks in less sensitive settings

Change in the Market Return	β	Expected Change in Firm's Stock Returns
$\pm 1\%$	0.0	$\pm 0.0\%$
$\pm 1\%$	0.5	$\pm 0.5\%$
$\pm 1\%$	1.0	$\pm 1.0\%$
$\pm 1\%$	1.5	$\pm 1.5\%$

The Security Market Line

An investment's risk premium is equal to beta times the risk premium of the market:

$$\beta_i = \frac{\text{Corr}(R_i, R_{Mkt}) \frac{\text{Std}(R_i)}{\text{Std}(R_{Mkt})}}{\frac{\text{Cov}(R_i, R_{Mkt})}{\text{Var}(R_{Mkt})}} = \frac{\text{Cov}(R_i, R_{Mkt})}{\text{Var}(R_{Mkt})}$$

$$r_i = r_f + \beta_i \times [\bar{R}_{Mkt} - r_f]$$

Beta is the OLS regression coefficient

An investment's risk premium is equal to beta times the risk premium of the market
Higher return to invest in risky securities

CAPM: The final frontier

$$E[R_i] = r_f + \beta_{i,market} * (E[R_{market}] - r_f)$$

Three key components:

1. The risk free (r_f)
2. The project specific (project i) beta $\beta_{i,market}$
3. The expected excess return of the market portfolio or expected market risk premium ($E[R_{market}] - r_f$)

Lecture 11

Intuition: Beta measures market risk:

- Beta is the “sensitivity” of the stock to the overall market
- Intuition: the percent change we expect in the stock for a one percent change in the value of the overall market

Discount rates for a project

How can we estimate the beta of a project? The previous slides emphasized equity betas. In general, the beta of a project/asset (firm) is not the beta of equity.

Equity is only one of the claims to the project, asset or firm!!!

Problem: unlike stock returns, the returns to real assets are not easily observable in the market: Equity risk is not the same as the risk of the assets: remember equity is a “residual” claim.

Yet, we can use equity betas to estimate the beta of all assets: the unlevered betas! Using the idea that the beta of a portfolio is the value weighted average of the betas of the individual assets in the portfolio.

Unlevered beta, free cash flow beta, asset beta 均 all mean the same.

Objective: measure the beta of a real investment

Simple case: an enterprise that is 100% equity financed with no excess cash. Assume equity betas are available (i.e. the equity trades) Then, I can use the estimate of the beta of the equity to approximate the beta of the asset!

Why? The returns received by equity investments ARE the returns derived from the real assets. Estimate the beta of the claims (in this case equity only) to get the beta of the assets!

Estimating the Beta of Real Assets Market Value Balance Sheet

Market Value of Assets $r_{\text{Unlevered}}$	Market Value of Debt r_{Debt}
	Market Value of Equity r_{Equity}

We can treat the assets as a portfolio of the debt and equity securities:

$$r_{\text{Unlevered}} = \frac{D}{D+E} r_{\text{Debt}} + \frac{E}{D+E} r_{\text{Equity}}$$

Which implies the following equation for betas:

$$\beta_{\text{Unlevered}} = \frac{D}{D+E} \beta_{\text{Debt}} + \frac{E}{D+E} \beta_{\text{Equity}} \quad \text{this is the free cashflow beta}$$

How do we determine project betas?

1. Identify comparable firms that are likely to have cash flow streams that have similar risk: Firms where the riskiness of the cash flow stream is the same as the project you are analyzing: the cash flows to the firm, revenue, etc

Always use comparable firms even if you have the betas for the firm you are valuing: betas are poorly estimated at best

2. Equity beta estimates for each comparable firm: Estimate your own equity betas. Measuring the slope of the line of best fit between the stock's and the S&P 500 returns

Regression coefficient: estimated using weekly/monthly data over 2-5 years

Use estimated equity betas. Beta data providers. Free online: Value Line, Yahoo / Google. Others: Bloomberg, Reuters, etc.

3. Estimate debt betas for each comparable firm: Debt betas may not be available: debt may not trade (returns are not available) Common assumptions:

- Debt is risk free: debt beta of zero if leverage is low
- Debt betas between 0.1 and 0.3: historical corporate debt returns

4. Estimate the market values of debt (D) and equity (E) used by each comparable using market values:

- Use market value of debt, net of excess cash. Include all interest-bearing debt
- Market value of debt: often approximated by its book value
- Market value of equity: share price times the number of shares outstanding. Never use book values!

5. Obtain the unlevered beta for each of the comparable firms:

$$\beta_{\text{Unlevered}} = \frac{D}{D+E} \beta_{\text{Debt}} + \frac{E}{D+E} \beta_{\text{Equity}}$$

6. Use the average unlevered beta in the CAPM equation:

$$E(r_{unlevered}) = r_f + \beta_{unlevered} * (E(R_{market}) - r_f)$$

AMERITRADE CASE

We expect that Ameritrade is a high beta company since people do more trading when the market is going well and less trading when the market is going bad.

Evaluating the investment

The firm's enterprise value is the present value of future cash flows:

$$EV = \frac{E[C_1]}{1+r_A} + \frac{E[C_2]}{(1+r_A)^2} + \frac{E[C_3]}{(1+r_A)^3} + \dots$$

r_A : Ameritrade discount rate

How do we assess the correct required return for projects? CAPM!

$$E(r_{Ameritrade}) = r_f + \beta_{Ameritrade} * (E(R_{market}) - r_f)$$

Risk-free Rate

Risk-free Interest Rate: What rate should we use to discount risk-free cash flows? We ALWAYS use current treasury rates.

Choosing the risk-free rate:

- Use the prevailing market yields, not historical rates. Prevailing yields reflect current market conditions: investors' opportunities
- Ideal: Match to the economic life of the project. Use the risk-free yield (from STRIPS prices) associated with the horizon of each cash flow. Problem: We don't have detailed cash flow projections.
- Practice: use the yield on a long term coupon bond whose maturity roughly matches the average horizon of the project. Ameritrade's investments are likely to be long-lived. Generally: use the yield on the 10 or 30 year coupon bond
- A reasonable approach would be to use the current yield on long-term US government bonds: 6.61% in Exhibit 3

Market Index

Some available indices: Dow Jones Industrial Average, S&P 500, Russell 2000, NASDAQ Composite

In the case:

- Should we use the Value Weighted NYSE/AMEX/NASDAQ index?
- Should we use the Equal Weighted NYSE/AMEX/NASDAQ index?

Theory suggests that we should use as broad an index as possible. The index should reflect all risky investment opportunities in the economy. In practice, most people use a broad stock market value weighted index: S&P 500

Finding beta

1. Find comparables
2. Estimate equity betas

Firm Name	Equity Beta	Comparable?
AG Edwards	1.45	
Bear Stearns	1.69	
Charles Schwab	2.30	Yes
Merrill Lynch	1.82	
Morgan Stanley Dean Witter	2.09	
PaineWebber	1.76	
Quick & Reilly	2.20	Yes
Raymond James	1.45	
Waterhouse	3.19	Yes

3. Estimate debt betas (assume $\beta_{\text{debt}}=0$ for now!)

4. Estimate the market value of debt and equity $D/(D+E)$, $E/(D+E)$ (given Exh4)

5. Find unlevered asset beta for each comparable

$$\beta_{\text{Unlevered}} = \frac{D}{D+E} \beta_{\text{Debt}} + \frac{E}{D+E} \beta_{\text{Equity}} \quad E(r_{\text{Ameritrade}}) = r_f + \beta_{\text{unleveredcomps}} * (E(R_{\text{market}}) - r_f)$$

Firm Name	Equity Beta	Comparable?	Debt/Value	Equity/Value	Unlevered Beta
AG Edwards	1.45				
Bear Stearns	1.69				
Charles Schwab	2.30	Yes	8.0%	92.0%	2.12
Merrill Lynch	1.82				
Morgan Stanley Dean Witter	2.09				
PaineWebber	1.76				
Quick & Reilly	2.20	Yes	0.0%	100.0%	2.20
Raymond James	1.45				
Waterhouse	3.19	Yes	38.0%	62.0%	1.98
			$\beta_{\text{Unlevered}}$	Average	2.10

Equity beta reflects the leverage of the company, if the company has a higher Debt/value, the equity beta is going to be higher.

6. Compute the average and use the CAPM!

$$E(r_{\text{Ameritrade}}) = r_f + \beta_{\text{Ameritrade}} * (E(R_{\text{market}}) - r_f)$$

$$E(r_{\text{Ameritrade}}) = r_f + 2.1 * (E(R_{\text{market}}) - r_f)$$

$$E(r_{\text{Ameritrade}}) = 0.0661 + 2.1 * (E(R_{\text{market}}) - r_f)$$

$$E(r_{\text{Ameritrade}}) = 0.0661 + 2.1 * 0.065$$

$$E(r_{\text{Ameritrade}}) = 20.2\%$$

Market risk premium = $E(R_{\text{market}}) - r_f$ The market offers a 6.5% higher expected return than the risk free.

The Capital Structure Question

So far we tried to answer the following question: Which projects should the firm invest in to maximize value?

Key tools: Cash-flows: FCF, NPV, etc. Discount rates: beta, CAPM, NPV

The next question: How should the firm raise the capital necessary to make these investments?

Should the firm: Borrow (issue debt)?, Sell shares (issue equity)?

Assets	Liabilities
Projects & Investments undertaken by firm	Securities issued by firm:
(Take Positive NPV Projects)	Debt
	Equity

Leverage and the Cost of Capital

Most large corporations use a mix of debt and equity financing

If a firm borrows, or issues stock, investors will demand a risk premium to compensate them for the risk they are taking. The risk, and therefore the risk premium, will vary by security

The firm's overall cost of capital is the weighted-average cost of all sources of funding

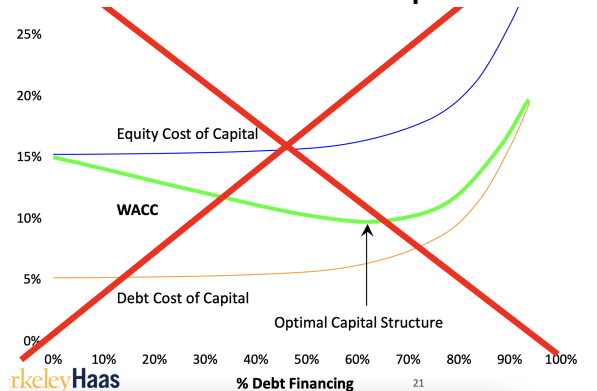
Weighted-Average Cost of Capital or WACC

$$WACC = \left(\frac{\% \text{ of Equity}}{\text{Financing}} \right) \times \left(\frac{\text{Equity Cost}}{\text{of Capital}} \right) + \left(\frac{\% \text{ of Debt}}{\text{Financing}} \right) \times \left(\frac{\text{Debt Cost}}{\text{of Capital}} \right)$$

Optimal Capital Structure: Objective: find the optimal (%E), and (%D) in the WACC. Minimize the WACC, maximize value!

Consider the following: The cost of debt financing is lower than the cost of equity financing. Does this imply that debt is better?

A Common View: Debt is Cheap



WRONG!!

Adding leverage increases the risk of equity. The initial cash flows go to debt holders: first to collect. As a result, both the volatility and the beta of equity increase

To compensate equity holders, the expected return on equity also increases!

The risk premium double when the risk double.

But, changes in the capital structure do not affect the WACC

- WACC of the **all-equity** firm:

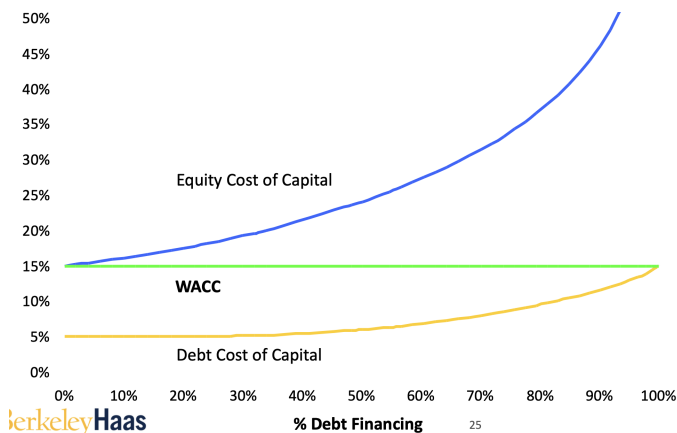
$$R_{WACC} = \frac{E}{D+E} R_E + \frac{D}{D+E} R_D = 1 \times 15\% + 0 \times 5\% = 15\%$$

- WACC of the **levered** firm:

$$R_{WACC} = \frac{E}{D+E} R_E + \frac{D}{D+E} R_D = \frac{1}{2} \times 25\% + \frac{1}{2} \times 5\% = 15\%$$

With no debt, the WACC is equal to the unlevered equity cost of capital. As the firm borrows at the low cost of capital for debt, its equity cost of capital rises: the net effect leaves the WACC unchanged.

M&M and WACC



You can't shift to a lower cost of debt to decrease the WACC because the equity cost of capital is going to increase to compensate equity holders for a higher risk.

How you purchase something doesn't affect its value. The pizza is the same!

Lessons

- Leverage increases the risk and the required return of equity. Debt is paid first and thus has less (or no) risk. Equity has "more concentrated" risk

- In a perfect capital market: Adding leverage does not change the firm's WACC. Even though debt is "cheaper," the cost of equity goes up by enough to eliminate any benefit. Not surprising: the risk of the project (business) did not change. Adding leverage does not change the total value of the firm

$$V(\text{Unlevered}) = V(\text{Debt}) + V(\text{Equity})$$

Whole = Sum of the parts

- Modigliani - Miller (M&M) Theorem. Financial policy is irrelevant: Any capital structure is as good as any other!

If two investments (one with leverage, one without) provide the same cash flows in every scenario, then they must sell for the same price

Value is determined by cash-flows NOT by how these cash-flows are sliced

Financial Policy "Irrelevance"

M&M implies that corporate financial policy is irrelevant to firm value. Capital structure does not affect the amount of capital the firm can raise. Trade in other market securities also does not matter

$$\text{NPV}(\text{Buying a Security}) = -\text{Price} + \text{PV}(\text{Cash Flows}) = 0$$

Key Assumption: Perfect Capital Markets

- No Taxes, no Bankruptcy Costs: interests are tax deductible and dividends aren't. Bankruptcy has cashflows consequences (e.g. legal costs) and indirect consequences (turnover management, changes in accs payables less days from our providers, reputation)
- No Agency Costs: conflicts between management and shareholders
- No Asymmetric Information
- Securities are fairly priced

Key Insight: To create value through financial policy, one needs to focus on a "market imperfection" that can affect cash flows

Is capital structure really irrelevant?

No! M&M simplifying assumptions do not hold in practice. Therefore: capital structure does matter in the real world!

M&M is still extremely useful as it tells us that if capital structure creates value then it must be because one of M&M's assumptions is not valid: i.e., it tells us where to look!

Examples of changes in leverage affecting cash-flows:

- Different tax treatment of debt and equity
- Costs of financial distress
- Incentives and agency costs
- Information asymmetry

Optimal capital structure: how to tradeoff the advantages and disadvantages of debt financing

Lecture 12

Leverage and Taxes

MM relied on perfect or frictionless capital markets. MM: any capital structure is as good as any other! Thus, capital structure must matter due to some market imperfection:

- Corporate income taxes
- Bankruptcy costs
- Agency costs (incentives)
- Differences in information
- Security mispricing

Corporate income taxes: Can leverage be used to reduce the corporate income taxes that the firm must pay, and thereby increase its value for investors?

Interest expenses deducted before tax

No corresponding deduction for dividends or share repurchases

Gain from Leverage

- **With vs. Without Leverage (2013)**

	<u>All Equity Firm</u>	<u>Levered Firm</u>
EBIT	\$1000	\$1000
Interest Paid to Debt Holders	\$0	\$100
Pre-Tax Income	\$1000	\$900
Tax at 35%	\$350	\$315
Net Income to Equity	\$650	\$585
	<u>All Equity Firm</u>	<u>Levered Firm</u>
	\$650	\$585 (equity)
		\$100 (debt)
		\$685

- What is the **benefit**?

- Total income to **all** investors
- Gain from leverage = $685 - 650 = \$35$

- Tax savings = $350 - 315 = \$35$

The Interest Tax Shield

Debt gives the firm an “interest tax shield” which reduces taxes each year:

Tax reduction = Corporate Tax Rate x Interest Payments

This tax shield:

- Reduces the taxes paid by the firm
- Increases the net-of tax cash-flows available to both debt and equity
- Increases the value of the firm: $PV(\text{future interest tax shields}) = T_c \times PV(\text{future interest payments})$

T_c : corporate tax rate

Value of the Interest Tax Shield

Special and simplest case:

Firm increases debt by D permanently: perpetually rolled over

Each period, the tax shield is $= T_c \times D \times r_D$. Given that D is permanent, the present value of tax shield is:

$PV(\text{ITS}) = (T_c \times D \times r_D) / r_D = T_c \times D$

- **Gain from leverage:** $V(\text{Levered Firm}) = V(\text{Unlevered}) + T_c \times D$

- **Change in value:** $\frac{V(\text{Levered Firm})}{V(\text{Unlevered})} = 1 + \frac{T_c \times D}{V(\text{Unlev})}$

Are these effects meaningful?

If $T_c = 35\%$:

for $D / V(\text{Unlev}) = 20\%$, firm value increases by about 7%

for $D / V(\text{Unlev}) = 50\%$, firm value increases by about 17.5%

Bottom line: debt tax shields matter!

Direct and indirect costs of financial distress

If debt tax savings are so large what limits debt use?

Direct bankruptcy costs:

- Cost of the legal proceedings around reorganization or liquidation
- Any costs directly related to the event of bankruptcy:
- Legal fees
- Accounting fees
- Advisory fees

Indirect costs of financial distress:

- Loss of (or damage to) intangibles, such as brands
- Difficulty retaining valuable employees
- Difficulty of maintaining relationships with customers and suppliers
- Distortions to investment behavior when firm is close to default
- Distortions to managers' incentives when firm is close to default

Trade-Off theory of capital structure

Firms trade off tax benefits of debt against bankruptcy and financial distress costs

Optimal leverage ratio is determined by:

- a. The value and probability of using tax shields, and
- b. The costs and probability of financial distress

Characteristic	Effect on Optimal Leverage
Profitability	Positive
Non-debt tax shields (eg. Depreciation)	Negative
Tangibility of assets	Positive
Volatility of cash flows	Negative
Size	Positive
Indirect costs of financial distress *	Negative

* Customer confidence, labor force might leave, suppliers won't ship, etc)

Weighted Average Cost of Capital

- The After-Tax Cost of Debt
- Each \$1 of interest paid gives the firm a \$0.35 tax benefit • Net after-tax cost of paying \$1 in interest is only \$0.65!
- Effective after-tax interest rate on debt = $r_D (1 - T_c)$

Pretax WACC:

$$r_{WACC} = \frac{E}{D+E} r_E + \frac{D}{D+E} r_D = r_u$$

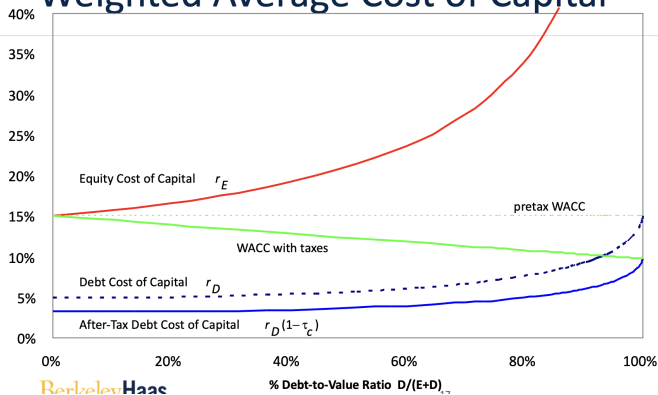
☐ Unchanged by leverage ratio

WACC with Taxes:

$$r_{WACC} = \frac{E}{D+E} r_E + \frac{D}{D+E} r_D (1 - T_c) = r_u - \frac{D}{D+E} T_c r_D$$

☐ Decreasing with leverage

Weighted Average Cost of Capital



Example: Diptron Inc's expansion

A manufacturer of electronic switches is evaluating an expansion:

- \$60 million expansion
- Expected to increase its FCF by \$7.5 m the first year, with 4% growth thereafter
- Diptron's tax rate is 40%
- The debt-equity ratio is 1/3 or $D/(D+E)=25\%$
- The equity cost of capital is 14.33%, and its cost of debt is 5% (pre-tax)

Questions:

1. What is the WACC?
2. What is the NPV of this project?

WACC: get shares of D and E

$$\begin{aligned} r_{WACC} &= \left(\frac{E}{V}\right)r_E + \left(\frac{D}{V}\right)r_D(1 - T_c) \\ &= 75\% \times 14\frac{1}{3}\% + 25\% \times 5\% \times (1 - 40\%) \\ &= 11.5\% \end{aligned}$$

Valuing the project

Enterprise value (EV) of the expansion: $V_L = 7.5 / (11.5\% - 4\%) = \100 million

NPV = -60 + 100 = \$40 million

WACC Method: Assumptions

Assumptions:

- Risk: Expansion has similar risk to the rest of the firm
- Leverage: Diptron D/E = D/E project today and in the future

How much debt will Diptron use to fund the expansion?

Capital structure = 75% equity + 25% debt (market value)

25% x \$60 million investment = \$15 million

25% x \$40 million inc. in mkt value = \$10 million

Total new debt = \$25 million

Lecture 13

Equity Financing Terms

IMPORTANT: When a project is announced, the equity value of the firm immediately increases by the NPV!!!

The issuance price of the stock thus reflects the NPV of the project to prevent arbitrage opportunities.

The price of the stock jumps to its ultimate value upon announcement of the project, i.e. there is no further change

The value of the assets is the value of the CF that they are going to generate, DO NOT INCLUDE UPFRONT COST IN THE VALUE OF THE ASSETS.

If a project is going to be financed 20% with debt and 80% with equity, we need to sell stock to finance it.

- Initial equity value = 20 million shares x \$15 = \$300 million
- NPV of the project = \$15.89 million - \$10 million = \$5.89 million (**here we do include the upfront cost**)
- Issuance Price = \$305.89 million / 20 million shares = \$15.29
- Number of shares issued:

$$\frac{\text{Upfront Cost} - \text{Debt Financing}}{\text{Share Price}} = \frac{10 - 3.18}{15.29} = 0.44$$

Recapitalization: issue debt to repurchase shares, to increase the benefits from the tax shield of the interests.

Share price changes upon announcement of a project reflecting its NPV (to prevent arbitrage opportunities during the deal), but then it remains flat. People think that when you repurchase shares, the price has to change but it is not true. It only changes when a project is announced.

Controlling Overinvestment: Benefit

- When cash is abundant, managers may tend to overinvest. Managers may be “empire builders” and like to invest/dislike to disinvest. Even if there are no good investment opportunities available, managers may:
 - Initiate negative NPV projects, or continue pet projects or failed investments

- Over-hire, pay excessive compensation, and consume lavish perks
- Make non-strategic acquisitions, or overpay for acquisitions

Overinvestment is wasteful and decreases the value of the firm

- Debt is a “disciplinary device” (think Private Equity 7 LOB): High leverage can force managers to payout cash rather than overinvest

Takeaway: Debt as disciplinary device. Low growth, high cash flow firms should have high leverage to maintain incentives for efficiency and reduce overinvestment.

When you are closed to bankruptcy, it would be profitable for equity holders to gamble for survive. They have nothing to lose and a lot to win. The people that get hurt are the debt holders. I would destroy value for the debt holders 7 Agency cost

Leverage and Risk-Taking: Agency Cost

Highly leveraged firms have an incentive to make very risky investments

Shareholders get the benefit if investment succeeds

Bondholders (debt) bear the loss if investment fails (limited liability)

Example:

Suppose firm has future FCF worth \$1,000 (current strategy)

New strategy worth \$1,300 (success) or \$300 (failure), equally likely (50,50)

Would you take it? (a) If the firm has no debt? (b) If firm has outstanding debt = \$900?

	Current Strategy	New Strategy		
		Success	Failure	Expected
Firm Value	1000	1300	300	800
Debt	900	900	300	600
Equity	100	400	0	200

Lecture 13

CAPM and Costs of Capital / Discount Rates

- Equity Cost of Capital / Discount Rate:

$$r_E = r_f + \beta_E(r_M - r_f)$$

- Debt Cost of Capital / Discount Rate:

$$r_D = r_f + \beta_D(r_M - r_f)$$

- Pre-tax WACC:

$$r_u = \left(\frac{E}{D+E}\right)r_E + \left(\frac{D}{D+E}\right)r_D = r_f + \beta_u(r_M - r_f)$$

- After-tax WACC:

$$r_{WACC} = \left(\frac{E}{D+E}\right)r_E + \left(\frac{D}{D+E}\right)r_D(1 - T_C) = r_u - \left(\frac{D}{D+E}\right)r_D T_C$$

Debt Overhang and Underinvestment

Highly levered firms may under-invest due to a “debt overhang”. When leverage is risky (not always paid), debt will be worth less than comparable risk-free debt (the higher its risk, the lower its price). New positive NPV investments may decrease the risk of default, and increase the value of the debt. Because some of the gains will accrue to debt holders, equity holders have less of an incentive to fund such new investments

Example:

- Suppose the market value of firm assets falls below the face value of debt
- Will equity holders contribute \$100 to fund new investments worth \$200?

	Book Value	Market Value	w/ New Investment	
Assets	1,000	750	950	Debt holders gain \$150
Debt	900	750	900	
Equity	100	0	50	Equity holders don't recoup their investment

n:

whose value depends on future growth should limit their leverage

Conclusion:

- Firms whose value depends on future growth should limit their leverage
- Low growth, mature firms can sustain more debt

Class Recap

Free cash flow

Think incrementally: compare the world if you take the project against the world if you don't
Count only cash flows that are caused by the project you are evaluating, but count all of them
Always ask: would a cash flow still occur even without the project? If yes, it is not incremental to your decision, and you should ignore it

- Ignore sunk costs
- Ignore any other cash flows that your decision does not affect
- Account for cash flow effects of your project elsewhere in the firm
- Account for changes in the timing of cash flows
- Pay special attention to the cash flows at the end of your project
- Abandonment costs, liquidation proceeds, etc.

Valuation

To value a firm or project

- Determine its FCF
- Make the FCF comparable by calculating their present values
- Compute the present values using your opportunity cost of capital

The two main drivers of firm (project) values are:

- FCF and their projected growth path
- Cost of capital

Optimizing your portfolio

To build an optimal portfolio, you have to forecast both

- Expected returns
- Risk

If you don't want to do this, you can simply buy the market and earn the market risk premium. Arguably a good idea for most investors

Diversification is crucial: eliminate lots of risk at no cost. Diversification eliminates firm-specific risk but does not eliminate systematic (or correlated) risk. Hence, investors expect to be compensated for holding systematic risk. The market does not reward you for holding risk that you could easily get rid of through diversification

Cost of capital (for a firm or project)

The return that investors require to be willing to hold the firm (project) in their portfolio

In practice, determined as an "opportunity cost of capital": What investors could earn by investing into an equivalent marketable security with the same risk as the firm (project)

The right measure of risk is again the covariance (beta) of the firm (project) with investors' portfolios
The CAPM assumes that all investors hold the market. This makes the covariance (beta) with the market the right measure of risk for any investment

Funding your firm (or project)

Firm value is mostly determined by the cash flows produced in the firm's operations. All we do on the right-hand side of the balance sheet is selling these cash flows off to different investors. This makes it difficult to create significant value on the right-hand side of the balance sheet

If there are no frictions, then it is impossible to create value on the right-hand side of the balance sheet (MM Theorem)

- No Taxes, No Bankruptcy Costs
- No Agency Costs
- No Asymmetric Information
- Securities are fairly priced

In the real world, there are frictions. Sometimes they are large and important, sometimes not
But the M&M intuition continues to hold: Capital structure affects value if and only if it affects cash flows. To increase firm value through a capital structure move, you need a cash flow story. This cash flow story has to be based on exploiting a market imperfection

M&M is an incredibly useful roadmap for creating value on the right-hand side of the balance sheet. It also protects us against bogus (i.e., not friction-based) motivations for capital structure moves

The (extended) trade-off theory

Trade-off theory of capital structure: Firms should choose an optimal level of debt financing that balances:

1. Tax benefits of using debt financing
2. Agency benefits of debt (e.g., disciplining managers)
3. Low asymmetric information costs of debt

Against:

1. Expected costs of financial distress and
2. Agency costs of debt (e.g., excessive risk-taking, debt overhang)

Optimal Capital Structure: Increase leverage as much as possible to capture tax benefits, but not so much as to incur large distress or agency costs